Kourovka Notebook Problem 21.24

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We prove the following theorem.

Theorem 1. *Let G be a finite group. If $\mathcal{P}(G)$ is a cograph, then $\mathcal{P}(G)$ is chordal.*

This answers a question of Bubboloni–Fumagalli–Praeger in the affirmative. We prove two lemmas which combined yield the result.

Lemma 2. *Let $n \geq 1$ and let C_n be cyclic of order n. Then $\mathcal{P}(C_n)$ is P_4–free if and only if $n = 1$, n is a prime power, or $n = pq$ with $p \neq q$ primes.*

Proof. In a cyclic group, for $x, y \in C_n$ we have $x \sim y$ in $\mathcal{P}(C_n)$ if and only if $|x| \mid |y|$ or $|y| \mid |x|$, and C_n has an element of order m if and only if $m|n$.

If $n = p^k$, then every element order is a power of p, so any two are comparable by divisibility; hence $\mathcal{P}(C_n)$ is the complete graph K_n and therefore P_4–free.

If $n = pq$ with $p \neq q$, suppose for contradiction that $x_1 - x_2 - x_3 - x_4$ is an induced P_4. Since $x_1 \not\sim x_3$, the orders $|x_1|$ and $|x_3|$ are incomparable, forcing $\{|x_1|, |x_3|\} = \{p, q\}$. But then x_2 is adjacent to both x_1 and x_3, so $|x_2|$ is divisible by both p and q, hence $|x_2| = pq$. Now x_4 is adjacent to x_3, so $|x_4| \in \{1, pq, |x_3|\}$, and thus $x_2 \sim x_4$, a contradiction.

If n is divisible by p^2q with distinct primes p, q, pick elements $x_1, x_2, x_3, x_4 \in C_n$ of orders p^2, p, pq, q, respectively. Then $x_1 \sim x_2 \sim x_3 \sim x_4$, while $x_1 \not\sim x_3$, $x_1 \not\sim x_4$, $x_2 \not\sim x_4$, so $\{x_1, x_2, x_3, x_4\}$ induces a P_4. If n is divisible by three distinct primes p, q, r, pick orders p, pq, q, qr to get an induced P_4 similarly. This concludes the proof. □

Corollary 3. *Let G be a finite group such that $\mathcal{P}(G)$ is P_4–free. Then for every $g \in G$, the cyclic subgroup $\langle g \rangle$ has order 1, p^k, or pq with $p \neq q$ primes.*

Lemma 4. *Let G be a group and suppose $\mathcal{P}(G)$ contains an induced 4–cycle on vertices x_1, x_2, x_3, x_4 in cyclic order. Then,*

$$x_1, x_3 \in \langle x_2 \rangle \cap \langle x_4 \rangle \quad \text{or} \quad x_2, x_4 \in \langle x_1 \rangle \cap \langle x_3 \rangle.$$

Proof. Note that the groups $\langle x_i \rangle$ are all distinct, as the cycle is induced. For any edge $\{x, y\}$ in $\mathcal{P}(G)$, either $\langle x \rangle \subsetneq \langle y \rangle$ or $\langle y \rangle \subsetneq \langle x \rangle$. Orient each cycle edge toward the endpoint generating the larger cyclic subgroup. If two consecutive edges are oriented in the same direction, say $\langle x_1 \rangle \subseteq \langle x_2 \rangle \subseteq \langle x_3 \rangle$, then transitivity gives $\langle x_1 \rangle \subseteq \langle x_3 \rangle$, hence $x_1 \sim x_3$, contradicting that the 4–cycle is induced. Thus the orientations alternate, so a pair of opposite vertices have cyclic subgroups containing those of their neighbors, giving the desired outcome. □

Proof of Theorem 1. Since induced cycles of length at least five contain P_4, it suffices to consider induced 4–cycles. Assume for contradiction that $\mathcal{P}(G)$ contains an induced 4–cycle on vertices x_1, x_2, x_3, x_4 in cyclic order. By Lemma 4, we may assume $x_1, x_3 \in H$, where $H = \langle x_2 \rangle \cap \langle x_4 \rangle$. The subgroup H is cyclic, since it is a subgroup of the cyclic group $\langle x_2 \rangle$. Since the 4–cycle is induced, $x_1 \not\sim x_3$ in $\mathcal{P}(G)$, and hence also in $\mathcal{P}(H)$. Thus H cannot have prime power order. By Corollary 3, $|H| = pq$ for distinct primes p, q. Now $\langle x_2 \rangle$ is cyclic and contains H, so $|x_2|$ is divisible by pq. Applying Corollary 3 to $\langle x_2 \rangle$, we conclude $|x_2| = pq$,

hence $\langle x_2 \rangle = H$. Similarly $\langle x_4 \rangle = H$. Therefore $\langle x_2 \rangle = \langle x_4 \rangle$, which implies $x_2 \sim x_4$ in $\mathcal{P}(G)$, contradicting that the 4–cycle on x_1, x_2, x_3, x_4 is induced. Hence $\mathcal{P}(G)$ has no induced 4–cycle, as desired. □

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Formal verification. The complete formal proof in Lean 4 can be explored and verified in an interactive session in the Lean 4 Web Editor without any local installation.